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Statistics for Business Administration

Exam: MA9712 / Practice Exam B

Date: Wednesday 5th August, 2026

Examiner: Prof. Mathias Drton

Time: 14:00 – 15:30

Working instructions

- This exam consists of pages with a total of 9 problems.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is **52 credits**.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one scientific pocket calculator
 - the cheat sheet, which is contained in this exam on page 16
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____

Early submission at _____

Problem 1

7.5 credits

A university surveys $n = 600$ students about their satisfaction with campus dining (*Satisfied*, *Neutral*, *Dissatisfied*) and records each student's meal-plan type (*Full*, *Partial*, *None*). The observed counts are:

Meal plan	Satisfied	Neutral	Dissatisfied	Total
Full	150	60	30	240
Partial	110	70	30	210
None	40	50	60	150
Total	300	180	120	600

- Identify the response variable and the explanatory variable, and state the type (categorical/numerical) of each.
- The students were selected by a simple random sample of the university's student body, but were not randomly assigned to a meal-plan type. Can the results be generalized to the whole student body? Can a causal claim "meal plan affects satisfaction" be made? Justify both answers.
- Which single graphical display is most appropriate to compare the satisfaction distribution across the three meal-plan groups? Name it.
- Compute the joint probability that a randomly selected student has *no* meal plan and is *Dissatisfied*.
- Show that the probability that a randomly selected student is *Satisfied* or has a *Full* meal plan equals 0.650 (to three decimal places).
- Suppose 8 students are drawn independently from the population, in which the proportion of *Dissatisfied* students is $p = 0.20$. Let X be the number of *Dissatisfied* students among the 8. Compute $\mathbb{P}(X = 3)$. You may use the hints:

```
dbinom(3, size = 8, prob = 0.2) ## [1] 0.1468
pbinom(3, size = 8, prob = 0.2) ## [1] 0.9437
pbinom(2, size = 8, prob = 0.2) ## [1] 0.7969
```

Problem 2

6 credits

A consumer group measures the battery life (in hours, under a standardized test) of $n = 50$ smartphones of one model. The sample has mean $\bar{x} = 18.0$ hours and standard deviation $s = 4.0$ hours. A boxplot of the data shows the first quartile at $Q_1 = 15.5$ hours and the third quartile at $Q_3 = 21.0$ hours.

- Read the interquartile range (IQR) of battery life from the boxplot information.
- State the conditions required by the Central Limit Theorem so that $\bar{x}$ is approximately normally distributed, and argue whether they are satisfied here.
- A 90% confidence interval for the mean battery life μ is reported as $[17.07, 18.93]$ hours. Justify this interval by computing the standard error and margin of error. You may use $z_{0.95} = 1.645$.
- Interpret the confidence interval in context in one sentence.

Problem 3

4 credits

A rapid test for a medical condition is evaluated. In the target population the prevalence of the condition is 4%. The test has sensitivity 92% (probability of a positive result given the condition is present) and specificity 95% (probability of a negative result given the condition is absent).

- Define suitable events for “has the condition” and “tests positive”, and describe the probability tree (branches and their probabilities).
- Compute the overall probability that a randomly tested person receives a positive test result.
- Compute the positive predictive value, i.e. the probability that a person actually has the condition given that they tested positive. Round to three decimal places.

Problem 4

5 credits

A clinical trial randomly assigns 400 volunteers to receive either a new vaccine ($n_1 = 200$) or a placebo ($n_2 = 200$). A mild side effect is reported by 46 people in the vaccine group and by 28 people in the placebo group. Researchers ask whether the side-effect rate differs between the two treatments.

- Let p_1 and p_2 denote the true side-effect proportions in the vaccine and placebo populations. State the null and alternative hypotheses for a two-sided test.
- Compute the observed difference in sample proportions $\hat{p}_1 - \hat{p}_2$.
- To obtain a p -value, the treatment labels were randomly reshuffled 10,000 times and the difference recomputed each time, giving the null distribution summarized below. Report the p -value.

```
# number of reshuffles with |difference| >= observed difference
sum(abs(perm_diffs) >= abs(obs_diff))    ## [1] 214
length(perm_diffs)                      ## [1] 10000
```

- State your conclusion at significance level $\alpha = 0.05$ in context.

Problem 5

4 credits

For each of the following statements, decide whether it is **true** or **false** and *justify your answer*. Only correct justifications are awarded with points, not the decision (true/false).

- A p -value of 0.03 means that the probability the null hypothesis is true equals 0.03.
- If we increase the confidence level from 90% to 99% while keeping the same data, the confidence interval becomes wider.
- For a fixed confidence level, quadrupling the sample size halves the margin of error of a confidence interval for a mean.
- The success–failure condition $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$ is used to justify the normal approximation for inference about a proportion.

Problem 6

4 credits

A genetics experiment crosses two pea-plant lines. Mendelian theory predicts that the four phenotype classes (call them A, B, C, D) occur in the ratio 9 : 3 : 3 : 1. Among $n = 160$ offspring the observed counts are:

Phenotype	A	B	C	D	Total
Observed	100	26	28	6	160

Test at $\alpha = 0.05$ whether the data are consistent with the predicted 9 : 3 : 3 : 1 ratio.

- State the null and alternative hypotheses for the goodness-of-fit test.
- Compute the expected counts under H_0 and verify the condition that every expected count is at least 5.
- Compute the chi-square test statistic, state the degrees of freedom, and bracket the p -value using the hints below. Conclude in context.

```
qchisq(0.90, df = 3)  ## [1] 6.251
qchisq(0.95, df = 3)  ## [1] 7.815
qchisq(0.99, df = 3)  ## [1] 11.345
```

Problem 7

9 credits

An education researcher compares final-exam scores (points, out of 100) of students taught by three methods: *Lecture*, *Flipped*, and *Online*. Each group has $n_i = 7$ students ($N = 21$). The group summary is:

Method	n_i	mean $\bar{y}_i$	sd s_i
Lecture	7	72	$\sqrt{20}$
Flipped	7	81	$\sqrt{20}$
Online	7	75	$\sqrt{20}$

The grand mean is $\bar{y} = 76$.

- State the null and alternative hypotheses of the one-way ANOVA.
- State the three conditions required for ANOVA and comment briefly on each.
- Fill in the missing (NA) entries of the ANOVA table below.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
method	2	NA	NA	NA	NA
Residuals	18	360	NA		

- Carry out the F -test at $\alpha = 0.05$ and state your conclusion. You may use:

```
qf(0.95, df1 = 2, df2 = 18)  ## [1] 3.555
qf(0.99, df1 = 2, df2 = 18)  ## [1] 6.013
```

- Compute the proportion of variability in exam scores explained by teaching method ($R^2 = \text{SSG}/\text{SST}$).

- f) The three pairwise comparisons are carried out with Bonferroni-adjusted p -values shown below. State the Bonferroni-adjusted significance level and which pairs of methods differ significantly at family-wise level 0.05.

	diff	p.adj
Flipped-Lecture	9	0.00425
Flipped-Online	6	0.06555
Online-Lecture	3	0.67660

Problem 8

8.5 credits

A company models monthly salary (in thousands of EUR) of $n = 120$ employees using years of experience and gender (*Female* is the baseline; `genderMale` = 1 for male employees), including an interaction term. The fitted `lm()` summary is shown below.

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      32.00      1.20      ?    < 2e-16
experience         1.50      0.18      ?    < 2e-16
genderMale        3.00      1.60      ?         ?
experience:genderMale 0.80      0.30      ?         ?
---
Residual standard error on 116 degrees of freedom

```

- Write the fitted regression equation for a general employee (using the indicator `genderMale`).
- Simplify the fitted equation for *male* employees (`genderMale`= 1).
- Interpret the `experience` coefficient (1.50) for female employees in context.
- Interpret the interaction coefficient (0.80): how does the effect of one additional year of experience differ between male and female employees?
- Test at $\alpha = 0.05$ whether the interaction term is significant. Compute its t -value and use the hint below to obtain the p -value; then conclude.

```
| 2 * pt(2.667, df = 116, lower.tail = FALSE) ## [1] 0.00876
```

Problem 9

5 credits

A telecom company models customer `churn` (*yes/no*) using `tenure` (months), `monthly_charges` (EUR), and `contract_type` (baseline *Month-to-month*; levels *One year*, *Two year*). The fitted logistic-regression model is shown below.

```

glm(formula = churn ~ tenure + monthly_charges + contract_type,
     family = _____, data = customers)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    -0.5200     0.2100   -2.48    0.0131
tenure          -0.0450     0.0060  -7.50    < 2e-16
monthly_charges  0.0310     0.0040   7.75    < 2e-16
contract_typeOne year -0.9100     0.1500  -6.07    < 2e-16
contract_typeTwo year -1.7600     0.2200  -8.00    < 2e-16

```

- Fill in the blank in the `glm()` call so that a logistic regression is fitted.

- b) Interpret the `tenure` coefficient (-0.0450) as an odds ratio, in context.
- c) Using a 0.5 probability cut-off, the model produces the confusion matrix below (rows = prediction, columns = actual). Compute the sensitivity and the specificity.

	Actual: No	Actual: Yes
Pred: No	820	90
Pred: Yes	80	210

- d) The full model has an ROC area under the curve (AUC) of 0.861; a reduced model that drops `contract_type` has AUC 0.833. Which model discriminates churners from non-churners better, and why?

Cheat sheet

Exploring data. $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$, $s = \sqrt{s^2}$, $\text{IQR} = Q_3 - Q_1$, fences $Q_1 - 1.5 \text{IQR}$, $Q_3 + 1.5 \text{IQR}$.

$$r_{(x,y),n} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{s_x s_y}$$

Probability. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$; $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ if independent; $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$;
 $\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$.

$$\mathbb{E}(X) = \sum_i x_i \mathbb{P}(X = x_i) \text{ (discrete), } \mathbb{E}(X) = \int x f(x) dx \text{ (cont.)},$$

$$\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2, \quad \text{SD}(X) = \sqrt{\text{Var}(X)}, \quad \text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y), \quad Z = \frac{X - \mathbb{E}(X)}{\sqrt{\text{Var}(X)}}.$$

Distributions.

$$\text{Geometric: } \mathbb{P}(X = x) = (1-p)^{x-1}p, \quad \mathbb{E}(X) = \frac{1}{p}, \quad \text{SD}(X) = \sqrt{\frac{1-p}{p^2}}.$$

$$\text{Binomial: } \mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad \mathbb{E}(X) = np, \quad \text{SD}(X) = \sqrt{np(1-p)}.$$

$$\text{Neg. Binomial: } \mathbb{P}(X = x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, \quad x \geq k. \quad \text{Poisson: } \mathbb{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad \mathbb{E}(X) = \lambda, \quad \text{SD}(X) = \sqrt{\lambda}.$$

Inference — categorical.

$$1 \text{ prop. CI: } \hat{p} \pm z^* \text{SE}(\hat{p}), \quad \text{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}; \quad \text{test: } Z = \frac{\hat{p} - p_0}{\text{SE}_0}, \quad \text{SE}_0 = \sqrt{\frac{p_0(1-p_0)}{n}}.$$

$$2 \text{ prop. CI: } (\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}; \quad \text{test: } \text{SE}_0 = \sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}, \quad \hat{p}_{\text{pool}} = \frac{\hat{p}_1 n_1 + \hat{p}_2 n_2}{n_1 + n_2}.$$

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}, \quad \text{expected}_{ij} = \frac{(\text{row } i)(\text{col } j)}{\text{total}}, \quad df = k - 1 \text{ (GOF) or } (r - 1)(c - 1).$$

Inference — numerical.

$$1\text{-sample } t: \bar{x} \pm t_{df}^* \text{SE}(\bar{x}), \quad T_{df} = \frac{\bar{x} - \mu_0}{\text{SE}(\bar{x})}, \quad \text{SE}(\bar{x}) = \frac{s}{\sqrt{n}}, \quad df = n - 1.$$

$$2\text{-sample } t: (\bar{x}_1 - \bar{x}_2) \pm t_{df}^* \text{SE}, \quad T_{df} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\text{SE}}, \quad \text{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \quad df = \min(n_1 - 1, n_2 - 1).$$

$$\text{Sample size: two-sample } n \geq \frac{2\sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2; \quad \text{one-sample } n \geq \frac{\sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2, \quad \alpha^* \in \{\alpha, \alpha/2\}.$$

$$\text{ANOVA \& regression. } F = \frac{\text{MSG}}{\text{MSE}}, \quad df_1 = k - 1, \quad df_2 = n - k.$$

$$\text{SSG} = \sum_{i=1}^k n_i (\bar{y}_i - \bar{y})^2, \quad \text{SST} = \sum_{j=1}^n (y_j - \bar{y})^2, \quad \text{SSE} = \text{SST} - \text{SSG}.$$

$$\text{Post-hoc } T_{df} = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{\text{MSE}/n_i + \text{MSE}/n_j}}, \quad df = n - k; \quad \text{Bonferroni } \alpha^* = \alpha/K, \quad p_{\text{adj},i} = K p_i, \quad K = \binom{k}{2}.$$

$$\hat{y} = b_0 + b_1 x, \quad b_1 = r_{(x,y),n} \frac{s_y}{s_x}, \quad \hat{\sigma} = \sqrt{\frac{1}{n-2} \sum (y_i - \hat{y}_i)^2}, \quad T_{df} = \frac{b_1 - \beta_{1,0}}{\text{SE}_{b_1}}, \quad df = n - 2.$$

$$R^2 = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \quad R_{\text{adj}}^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2 / (n - k - 1)}{\sum (y_i - \bar{y})^2 / (n - 1)}, \quad \text{AIC} = c + n \ln(\sum e_i^2) + 2k.$$

Logistic regression. $\text{logit}(p_i) = \log \frac{p_i}{1-p_i} = \beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i}$, $\text{OR}_j = e^{\beta_j}$, sensitivity = $\mathbb{P}(\text{Test+} | \text{Cond+})$, specificity = $\mathbb{P}(\text{Test-} | \text{Cond-})$.

Quantile hints (standard normal). $z_{0.70} = 0.524$, $z_{0.80} = 0.842$, $z_{0.90} = 1.282$, $z_{0.95} = 1.645$, $z_{0.975} = 1.960$, $z_{0.99} = 2.326$, $z_{0.995} = 2.576$.